

PHYSICS PAPER 1

Form 3 — Forces & Newton's Laws — Exam Revision

Time Allowed: 75 minutes · Full Mark: 75 · This paper must be answered in English

Name: _____
Class: _____ ()

GENERAL INSTRUCTIONS

- There are **THREE sections**: A (True/False, 10 marks), B (Multiple Choice, 30 marks), C (Structured Questions, 35 marks).
- Section A: Mark '**A**' for TRUE and '**B**' for FALSE.
- Section B: Each question carries **2 marks**. Choose the best answer. No marks deducted for wrong answers.
- Section C: Show ALL working clearly. Take $g = 10 \text{ m s}^{-2}$ throughout.
- Diagrams are **NOT** necessarily drawn to scale.

Section A — True or False**10 marks**

Mark 'A' for a TRUE statement and 'B' for a FALSE statement. (1 mark each)

1. Weight is a vector quantity and always acts vertically downward.
2. The normal reaction force on an object on a horizontal surface is always equal to its weight.
3. According to Newton's First Law, a moving object will continue moving in a straight line at constant speed if no net force acts on it.
4. The net force on an object is zero if and only if the object is stationary.
5. Newton's Third Law states that action and reaction forces act on the same object.
6. If the mass of an object doubles and the net force acting on it also doubles, the acceleration remains unchanged.
7. Friction always acts in the opposite direction to the motion or intended motion of an object.
8. An object in free fall has a net force equal to its weight.
9. A heavier object always falls faster than a lighter one in a vacuum.
10. The tension in a string connecting two objects being pulled together is equal to the net force applied on the whole system.

END OF SECTION A**Section B — Multiple Choice****30 marks**

There are 15 questions. Choose the BEST answer. (2 marks each)

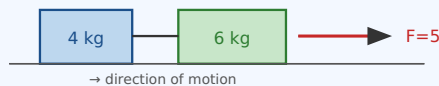
11. A book of mass 1.5 kg rests on a horizontal table. What is the normal reaction force from the table on the book?
 - A. 0 N
 - B. 1.5 N
 - C. 15 N
 - D. 150 N
12. A net force of 24 N acts on a 6 kg object. What is the acceleration produced?
 - A. 0.25 m s^{-2}

- B. 4 m s^{-2}
- C. 18 m s^{-2}
- D. 144 m s^{-2}

13. An object of mass 5 kg is pushed across a rough horizontal floor by a horizontal force of 40 N. The frictional force is 15 N. What is the acceleration of the object?

- A. 3 m s^{-2}
- B. 5 m s^{-2}
- C. 8 m s^{-2}
- D. 11 m s^{-2}

(For questions 14 and 15) A 4 kg block and a 6 kg block are connected by a light string and pulled along a smooth horizontal surface by a force $F = 50 \text{ N}$ as shown. The blocks move together.



14. What is the acceleration of the system?

- A. 5 m s^{-2}
- B. 8.3 m s^{-2}
- C. 10 m s^{-2}
- D. 12.5 m s^{-2}

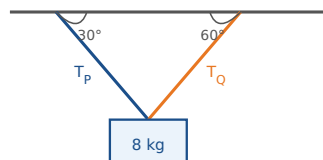
15. What is the tension in the string between the two blocks?

- A. 20 N
- B. 25 N
- C. 30 N
- D. 50 N

16. A man of mass 70 kg stands in a lift. The lift accelerates upward at 2 m s^{-2} . What is the normal reaction from the lift floor on the man?

- A. 560 N
- B. 700 N
- C. 840 N
- D. 1540 N

17. A box of mass 8 kg is suspended by two strings as shown. String P makes 30° with the ceiling and String Q makes 60° with the ceiling. Which statement about the tensions T_P and T_Q is correct?



- A. $T_P > T_Q$
- B. $T_P = T_Q$
- C. $T_P < T_Q$
- D. $T_P + T_Q = 80 \text{ N}$

18. A rocket of total mass 2000 kg produces a thrust of 30 000 N. Taking $g = 10 \text{ m s}^{-2}$, what is the initial acceleration of the rocket when launched vertically?

- A. 5 m s^{-2}

- B. 10 m s^{-2}
- C. 15 m s^{-2}
- D. 20 m s^{-2}

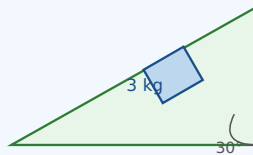
19. A car of mass 1200 kg decelerates from 20 m s^{-1} to rest in 8 seconds. What is the braking force acting on the car?

- A. 150 N
- B. 3000 N
- C. 6000 N
- D. $24\,000 \text{ N}$

20. A person pushes against a wall with a force of 60 N . According to Newton's Third Law, which statement correctly describes the reaction force?

- A. The wall pushes on the person with 60 N in the same direction.
- B. The wall pushes on the person with 60 N in the opposite direction.
- C. The wall pushes on the person with more than 60 N .
- D. There is no reaction if the wall does not move.

(For questions 21 and 22) A block of mass 3 kg is on a smooth inclined plane at 30° to the horizontal. The block is released from rest and slides down the plane. ($\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.866$, $g = 10 \text{ m s}^{-2}$)



21. What is the component of the weight of the block acting along the inclined plane (down the slope)?

- A. 15 N
- B. 25.98 N
- C. 30 N
- D. 51.96 N

22. What is the acceleration of the block down the inclined plane?

- A. 5 m s^{-2}
- B. 8.66 m s^{-2}
- C. 10 m s^{-2}
- D. 17.32 m s^{-2}

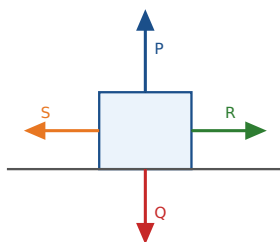
23. A person of mass 60 kg stands on a weighing scale inside a lift. The scale reads 48 kg . What is the acceleration of the lift?

- A. 2 m s^{-2} downward
- B. 2 m s^{-2} upward
- C. 0.8 m s^{-2} downward
- D. 10 m s^{-2} downward

24. Two forces of 6 N and 8 N act on an object at right angles to each other. What is the magnitude of the resultant force?

- A. 2 N
- B. 7 N
- C. 10 N
- D. 14 N

25. The figure below shows a free body diagram of a box being dragged at constant velocity across a rough floor. Which of the following correctly identifies ALL four forces?



- A. P = normal reaction, Q = weight, R = applied force, S = friction
- B. P = weight, Q = normal reaction, R = friction, S = applied force
- C. P = friction, Q = applied force, R = weight, S = normal reaction
- D. P = applied force, Q = friction, R = normal reaction, S = weight

END OF SECTION B

Show ALL working clearly. Marks are awarded for correct working even if the final answer is wrong. ($g = 10 \text{ m s}^{-2}$)

Question 1 (10 marks)

A crate of mass 20 kg rests on a rough horizontal floor. A horizontal force of 90 N is applied to push the crate. The coefficient of friction between the crate and the floor creates a frictional force of 30 N opposing motion.

- (a) Draw a fully labelled free body diagram of the crate, showing ALL forces acting on it. (4 marks)

Draw your free body diagram here

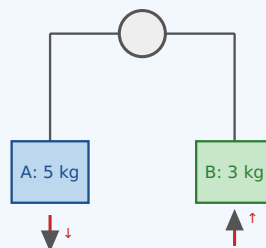
- (b) Calculate the net force acting on the crate. (2 marks)

- (c) Calculate the acceleration of the crate. (2 marks)

- (d) If the crate starts from rest, find the velocity of the crate after 4 seconds. (2 marks)

Question 2 (12 marks)

A 5 kg block (Block A) and a 3 kg block (Block B) are connected by a light inextensible string over a smooth, light pulley as shown (an Atwood machine). The blocks are released from rest. Ignore air resistance and the mass of the string.



- (a) State which block will move downward and explain why. (2 marks)

- (b) By applying Newton's Second Law to the whole system, calculate the acceleration of the blocks. (4 marks)

(c) Calculate the tension in the string. (3 marks)

(d) Find the speed of the blocks 2 seconds after they are released. (2 marks)

(e) If block B has a mass of 5 kg instead of 3 kg (equal to block A), state the acceleration and tension in the string. Explain your answer without calculation. (1 mark)

Question 3 (13 marks)

A car of mass 1500 kg is travelling at 30 m s^{-1} on a straight horizontal road. The driver sees an obstacle and applies the brakes. The car decelerates uniformly at 6 m s^{-2} until it stops.

(a) Calculate the braking force acting on the car. (2 marks)

(b) Calculate the time taken for the car to stop. (2 marks)

(c) Calculate the braking distance (distance travelled from when brakes are applied to when the car stops). (2 marks)

(d) On the same road, the car now travels down a slope inclined at 10° to the horizontal ($\sin 10^\circ \approx 0.174$). The same braking force is applied. Calculate the net force on the car and its new deceleration. (4 marks)

(e) Using your results, explain why cars take a longer distance to stop when travelling downhill compared to on a flat road, even with the same braking force applied. (3 marks)

ANSWER KEY — For Teacher Use Only

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Section A — True / False

- | | | | | |
|--|---|---|--|---|
| 1. A (True) | 2. B (False — only on a horizontal surface with no other vertical forces) | 3. A (True — Newton's 1st Law) | 4. B (False — zero net force → constant velocity, which includes moving) | 5. B (False — action & reaction act on DIFFERENT objects) |
| 6. A (True — $F=ma$; $2F/2m = a$, unchanged) | 7. A (True) | 8. A (True — only weight acts, net force = mg) | 9. B (False — in a vacuum all objects fall at the same rate) | 10. B (False — tension $\neq$ applied force; tension depends on mass of one block only) |

Section B — Multiple Choice

- | | | | | |
|---|--|---|--|---|
| 11. C ($N = mg = 1.5 \times 10 = 15 \text{ N}$) | 12. B ($a = F/m = 24/6 = 4 \text{ m s}^{-2}$) | 13. B ($F_{\text{net}} = 40 - 15 = 25 \text{ N}$; $a = 25/5 = 5 \text{ m s}^{-2}$) | 14. A ($a = 50/(4+6) = 5$) | 15. A ($T = m_A \times a = 4 \times 5 = 20 \text{ N}$) |
| 16. C ($N = m(g+a) = 70 \times (10+2) = 840 \text{ N}$) | 17. C ($T_P < T_Q$ — string at shallower angle to vertical has greater tension) | 18. A ($F_{\text{net}} = 30000 - 20000 = 10000 \text{ N}$; $a = 10000/2000 = 5 \text{ m s}^{-2}$) | 19. B ($a = 20/8 = 2.5 \text{ m s}^{-2}$; $F = 1200 \times 2.5 = 3000 \text{ N}$) | 20. B (Newton's 3rd Law: equal magnitude, opposite direction) |
| 21. A ($W \sin 30^\circ = 3 \times 10 \times 0.5 = 15 \text{ N}$) | 22. A ($a = g \sin 30^\circ = 10 \times 0.5 = 5 \text{ m s}^{-2}$) | 23. A (Scale reads $48 \text{ kg} \rightarrow N = 480 \text{ N}$; $N = m(g-a) \rightarrow 480 = 60(10-a) \rightarrow a = 2 \text{ m s}^{-2}$ downward) | 24. C ($\sqrt{6^2+8^2} = \sqrt{100} = 10$) | 25. A |

$\triangle$ Q17: The string making a smaller angle with the ceiling (P at 30°) is closer to horizontal, so its vertical component is smaller — meaning T_P must be smaller. $T_P \sin 30^\circ + T_Q \sin 60^\circ = 80 \text{ N}$; $T_P \cos 30^\circ = T_Q \cos 60^\circ \rightarrow T_P/T_Q = \cos 60^\circ / \cos 30^\circ = 0.5/0.866 \approx 0.577$, confirming $T_P < T_Q$. Answer C.

Section C — Structured Questions

Q1(a) Free body diagram: Four forces: Weight $W = 200 \text{ N}$ downward; Normal reaction $N = 200 \text{ N}$ upward; Applied force $F = 90 \text{ N}$ to the right; Friction $f = 30 \text{ N}$ to the left. (1 mark each correct force with label and direction)

Q1(b) $F_{\text{net}} = 90 - 30 = 60 \text{ N}$ (horizontal, in direction of motion)

Q1(c) $a = F_{\text{net}}/m = 60/20 = 3 \text{ m s}^{-2}$

Q1(d) $v = u + at = 0 + 3 \times 4 = 12 \text{ m s}^{-1}$

Q2(a) Block A (5 kg) moves downward because it has greater weight, so the net downward force on the system is due to the difference in weights: $5g - 3g = 20 \text{ N} > 0$.

Q2(b) Net force on system = $(m_A - m_B)g = (5-3) \times 10 = 20 \text{ N}$; Total mass = $5+3 = 8 \text{ kg}$; $a = 20/8 = 2.5 \text{ m s}^{-2}$

Q2(c) For Block B (going up): $T - m_B \cdot g = m_B \cdot a \rightarrow T = m_B(g+a) = 3 \times (10+2.5) = 37.5 \text{ N}$

Check: For Block A (going down): $m_A \cdot g - T = m_A \cdot a \rightarrow T = m_A(g-a) = 5 \times (10-2.5) = 37.5 \text{ N} \checkmark$

Q2(d) $v = u + at = 0 + 2.5 \times 2 = 5 \text{ m s}^{-1}$

Q2(e) Acceleration = 0 m s^{-2} ; Tension = $m_A \cdot g = 5 \times 10 = 50 \text{ N}$. Equal masses → equal weights → net force = 0 → no acceleration. Tension supports each block's full weight.

Q3(a) $F_{\text{braking}} = ma = 1500 \times 6 = 9000 \text{ N}$

Q3(b) $v = u + at \rightarrow 0 = 30 - 6t \rightarrow t = 5 \text{ s}$

Q3(c) $v^2 = u^2 + 2as \rightarrow 0 = 900 - 2 \times 6 \times s \rightarrow s = 75 \text{ m}$

Q3(d) Component of weight down the slope = $mg \sin 10^\circ = 1500 \times 10 \times 0.174 = 2610 \text{ N}$ (acts in same direction as motion, opposing braking).

Net force = $F_{\text{braking}} - mg \sin 10^\circ = 9000 - 2610 = 6390 \text{ N}$ (decelerating).

New deceleration $a = 6390/1500 = 4.26 \text{ m s}^{-2}$

Q3(e) When travelling downhill, a component of the car's weight acts in the direction of motion (down the slope), which

partially cancels the braking force. This reduces the net decelerating force on the car, resulting in a smaller deceleration. With a smaller deceleration, the car takes a longer time and distance to come to a complete stop compared to braking on a flat road where no component of weight opposes the braking force. (Award 1 mark per valid point, max 3.)